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Renewed Paleomagnetic Study of the Lisbon Volcanics and Implications for the Rotation of the Iberian Peninsula

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Two paleomagnetic studies of the Lisbon basalts by N. D. Watkins and A. Richardson and by R. Van der Voo bearing on the problem of the rotation of the Iberian Peninsula have lately been criticized with tectonic and paleomagnetic arguments. It was therefore concluded that a renewed study based on a new collection of samples might settle some points of discussion. As was reported previously, the remanent magnetizations of these Eocene lavas are of normal polarity only. An unfolding test proves the characteristic magnetizations to be pre-folding. From 176 samples, a mean direction of magnetization has been found, $D: 352°$, $I: +40°$, and $\alpha_{95}: 3°$, giving unit weight to sites ($N = 33$) in the statistical analysis. A comparison shows that there is such a slight difference between the paleomagnetic Late Cretaceous and Eocene data from Portugal and those from stable Europe, that any post-Eocene rotation of the Iberian Peninsula cannot have been more than a few degrees. This conclusion does not conflict with marine geophysical work reported so far.

In recent years the origin of the Bay of Biscay has appealed to paleomagnetism as a useful test of related theories. Paleomagnetic data from Pyrenean rocks [*Van der Lingen*, 1960; *Schwarz*, 1963; *Girdler*, 1965; *Van Dongen*, 1967] and from several formations of the Iberian Meseta [*Van der Voo*, 1967, 1969] are consistent with the hypothesis that the Bay of Biscay was created by rifting due to a post-Triassic anticlockwise rotation of the Iberian Peninsula, as *Carey* [1958] and *Bullard et al.* [1965] proposed. Paleomagnetic studies have also been made of Eocene volcanic rocks near Lisbon [*Watkins and Richardson*, 1968*a*; *Van der Voo*, 1968], but the validity of these results has been challenged [*Watkins and Richardson*, 1968*b*, 1971; *Storetvedt*, 1970]. The present study of 215 samples from 34 sites helps to resolve some of the questions raised by the earlier studies.

GEOLOGICAL SITUATION OF THE LISBON BASALTS AND SAMPLING DETAILS

The stratigraphical position of the basalt flows is between Upper Cretaceous limestone (Turonian) and the Oligocene Benfica formation; the basalt flows cover the limestone uncomformably. Clearly stratified clays, sands, and pyroclastic materials are intercalated between the flows. These intercalations contain Eocene freshwater molluscs (*Bulimus carnaxidensis, B. olisiponensis, Anadromus ribeiroi*), some vertebrates and plants (*Cinnamomum broteri*) [*De Jesus and Zbyszewski*, 1952; *Zbyszewski*, 1963, 1964]. These authors report more than six flows in a section west of Lisbon; northwest of Lisbon, we could distinguish at least eight flows. Poor outcrops, however, prevent correlations of the various sections.

The whole sequence of Upper Mesozoic, Eocene basalts, and Oligocene has been folded in a few large asymmetric anticlines and synclines, apparently owing to draping over faults in the basement. Differential vertical movements and tilting of blocks appear to be characteristic of the whole area. Figure 1 shows a geologic map after *Zbyszewski* [1963, 1964], and the block diagram of Figure 2 illustrates the folded structure northwest of Lisbon.

At most sampling sites, southeasterly dips between 5° and 20° prevailed, whereas at 12 sites northerly dips between 30° and 50° were observed. The dips were measured in the intercalated fine-grained freshwater deposits. The latter do not show any cross bedding or other

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signs of primary dip. They were deposited horizontally under quiet conditions in lakes or on flood plains. The present dips are due to later tectonic tilting.

All 215 samples (34 sites) were geographically oriented in situ with a sun compass.

ANALYSIS OF THE NRM

The magnetic properties will be discussed only briefly, since these have been described in detail by *Watkins and Richardson* [1968*a*].

The measurements of the NRM with astatic magnetometers revealed intensities between 10^{-3} and 2.10^{-2} emu/cm^3 and Q values generally higher than 1. Thermal and ac magnetic field demagnetizations of 187 samples were carried out with standard methods employed at Utrecht University [*As*, 1960; *Zijderveld*, 1967]. The remain-

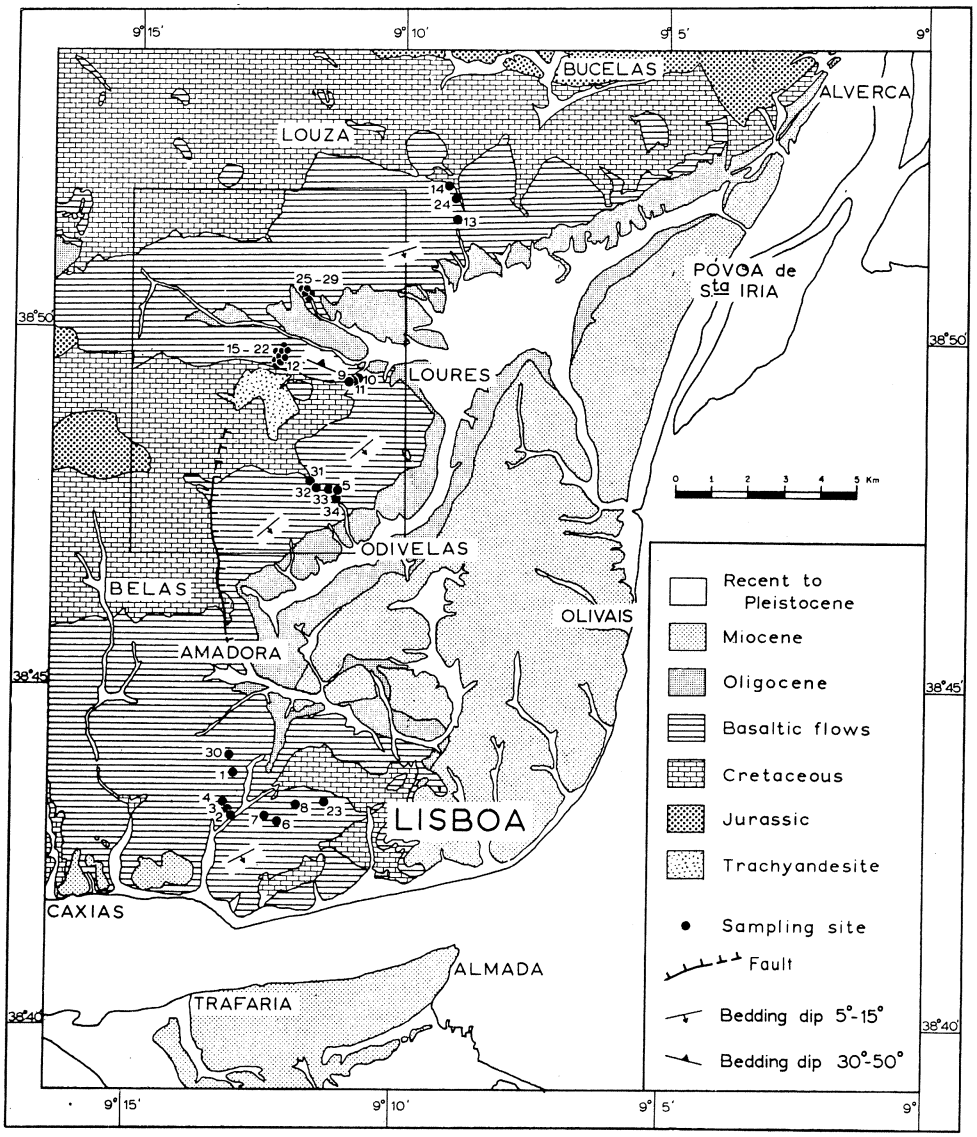
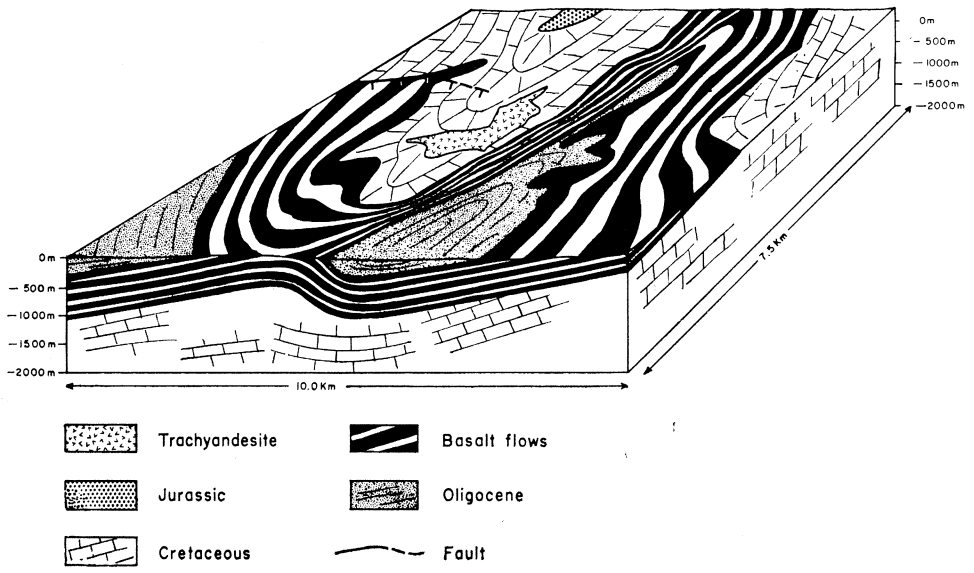


Fig. 1. Geologic map of the Lisbon area, after *Zbyszewski* [1963, 1964]. Sampling sites are shown. The rectangle is the outline of the block diagram of Figure 2.



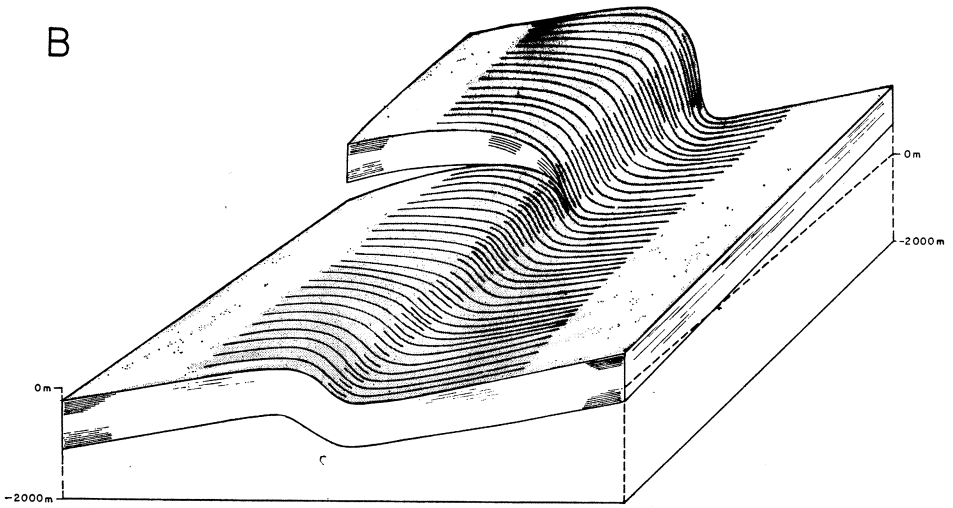


Fig. 2. (*A*) Block diagram of the Loures anticlinal structure northwest of Lisbon (rectangle in Figure 1) viewed from the east. (*B*) Idealized structure of the basalt sequence, before erosion started (same rectangle as *A*).

ing 28 samples have been stored for future experiments.

Demagnetization treatment generally revealed two magnetic constitutents (Figure 3). The 'softer' constituent in the samples, having aberrant directions, are considered to be of recent origin. The 'harder' constituents had directions, which, after unfolding, were similar for all samples, being thus characteristic for these Lisbon basalts.

The samples can be divided into two groups according to the ratio secondary/characteristic magnetizations and according to the 'hardness' of the characteristic magnetizations (or stability, as Watkins and Richardson prefer to call it). In one group of samples the secondary magnetizations constitute up to more than 80% of the total NRM (Figures 3*A*, *B*, *C*); the characteristic magnetizations of these samples were eliminated in the ac magnetic field trajectory between

200 and 1000 Oe, or in temperatures between 300° and 585°C. The other group of samples has either small secondary magnetizations or none at all (Figures 3*D*, *E*, *F*); the characteristic magnetizations of this group were destroyed in ac fields between 300 and 3000 Oe or higher, or in temperatures between 300° and 615° or higher. As *Watkins and Richardson* [1968*a*] pointed out in detail, this behavior can be attributed to a different oxidation state of the flows, or, as is now observed, a different oxidation state between various samples of one flow. There is, however, no distinction possible as far as the paleomagnetic directions are concerned.

The demagnetization diagrams of Figure 3 clearly demonstrate that there are no other magnetic constituents present in the samples, nor is there evidence of spurious components produced by demagnetization. For each sample, the direction of magnetization could be accurately determined from the demagnetization diagrams like those given in Figures 3 and 4, following the

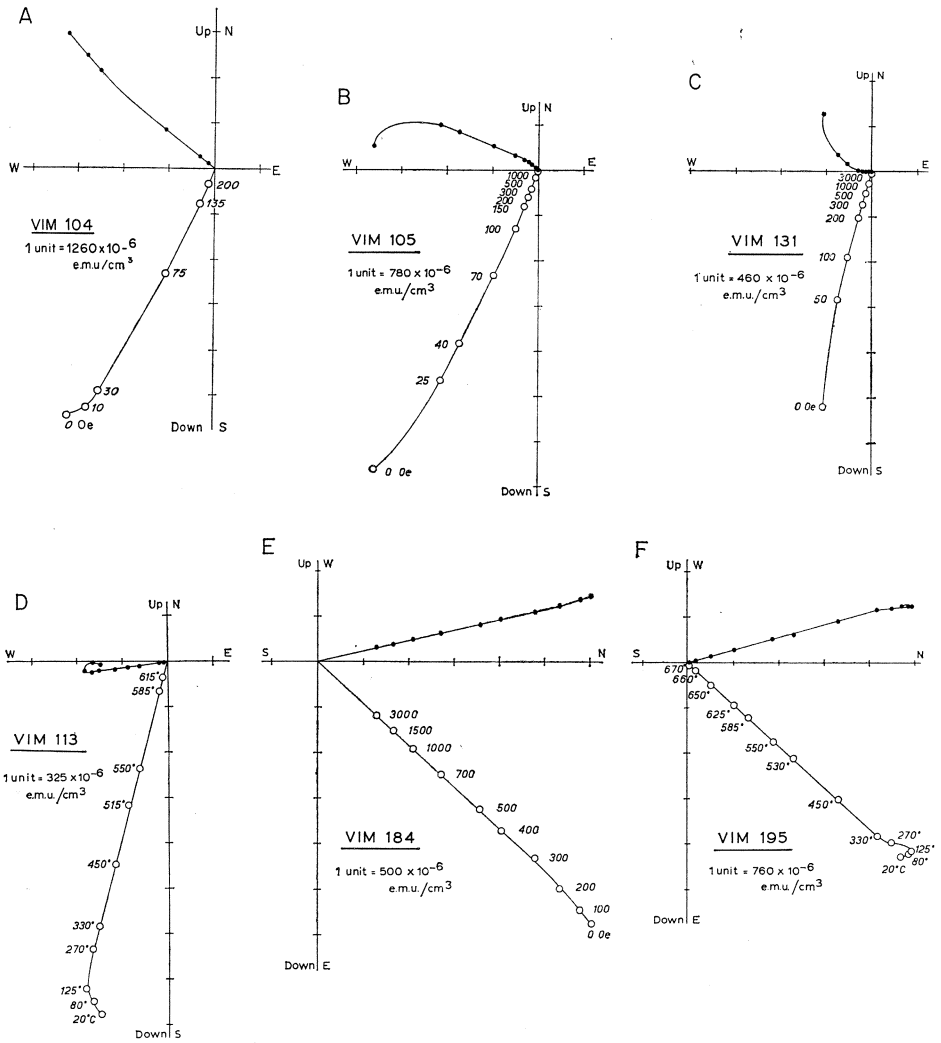


Fig. 3. Orthogonal projection of the successive end points of the vector of magnetization during progressive demagnetization. Solid symbols represent projections on the horizontal plane, open symbols on the north-south or east-west vertical plane. *A*, *B*, samples from site 11; *C*, sample from site 15; *D*, site 12; *E*, site 24; *F*, site 26.

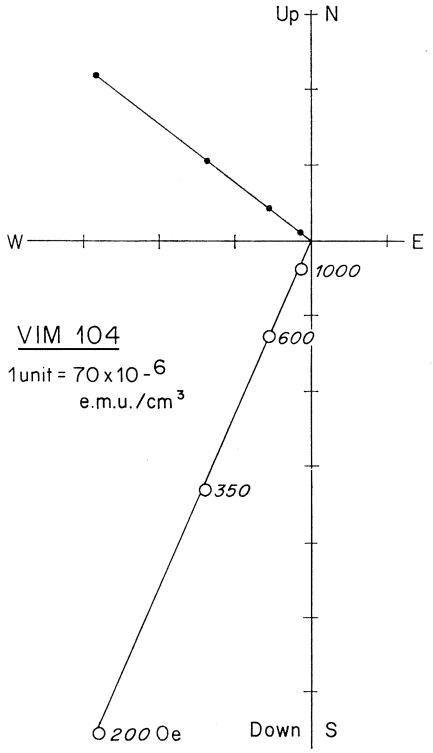


Fig. 4. Enlarged part of Figure 3A for the ac magnetic field trajectory between 200 and 1000 oe. For explanation of the symbols, see Figure 3.

method described by *Zijderveld* [1967], even in samples where only a small hard characteristic magnetization was present (Figure 4). Given these directions, there is no reason to invoke unrealistic computation methods for the site mean directions, as for example the method of minimum dispersion (see also *Storetvedt's* [1970] comments).

About 10 samples were not used for the computation of the mean directions of magnetization, because they had clearly aberrant directions of magnetization.

The directions of characteristic magnetization were without exception of normal polarity, as can be seen in Figure 5. The correction for the dip of the flows has been applied in Figure 5*B*. Comparison with Figure 5*A* reveals a marked decrease in the scatter of the site mean directions. We will discuss this fold test in the next section.

FOLD TEST AND MEAN DIRECTION OF MAGNETIZATION

One of the most important tests in paleomagnetism has always been the fold test [*Graham*, 1949], since it enables determination of a pre-folding age of the magnetization. The anti- and synclinal structure (see Figure 2) of the area northwest of Lisbon permits an unfolding test. As is shown in Figure 5, the magnetizations are definitely pre-folding. This is convincingly illustrated by the statistical parameters of *Fisher's* [1953] method. Before tectonic correction, these values were K: 15 and α_{95}: 6.5°, whereas after normal tectonic correction they became K:52 and α_{95}: 3.5° ($N = 33$).

In Figures 1 and 2 it can be seen that the Loures anticlinal structure northwest of Lisbon is asymmetrical and has an axial dip toward the

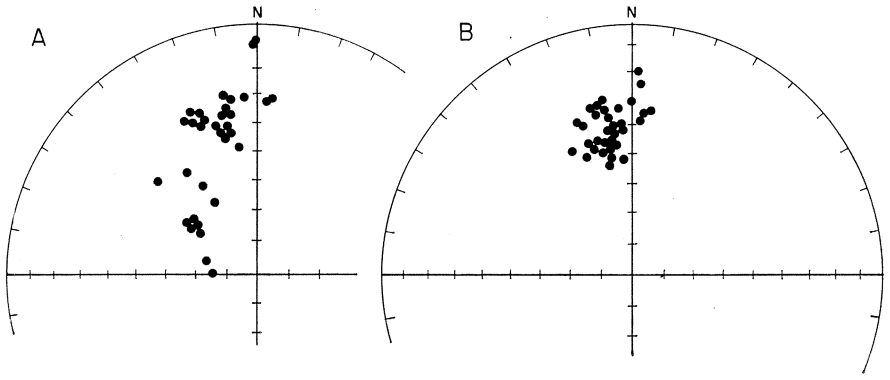


Fig. 5. Equal area projection of the site mean directions of characteristic magnetization. All directions are pointing downward. *A*, before tectonic correction; *B*, after tectonic correction.

east, which was established at between 15° and 16°. This geological situation points to a later southeastward tilting of a flexure initially trending east-west and dipping north. The northerly dipping flexure underwent a second tilting, as is evidenced by the southward dip of the southern flank of the resulting 'anticlinal' structure. When computing the site mean directions with respect to the (ancient) horizontal plane, we have therefore taken into account the effect of both the earlier flexure and the later tilting.

In doing so, a mean direction of magnetization with D:352°, I:+40°, K:66, and α_{95}:3° was computed from 33 site mean directions. The sites generally comprised 5 or 6 samples and yielded α_{95}'s between 2° and 9° (Table 1). An

TABLE 1. Summary of NRM and Site Mean Directions of Characteristic Magnetization of the Lisbon Basalts Obtained by Partial Progressive ac Demagnetization in the Trajectory between 300 and 1000 Oersteds

		Total NRM				Characteristic Directions of Magnetization: Before T.C.		After T.C.*				Strike and Dip of Bedding
No.	R_s	D	I	α_{95}	K	D	I	D	I	α_{95}	K	
1	5/7	220	+58	27	6	347	+36.5	352	+42.5	7.5	101	36/8
2	6/6	348	+32	6.5	105	340	+33	341.5	+37.5	4	260	45/5
3	6/6	12	+30	10.5	40	3	+32	5.5	+35.5	4.5	225	45/5
4	7/7	5	+37	7.5	65	4.5	+31	7	+34.5	3	385	45/5
5	4/5	359	+18	9.5	64	359.5	+ 9.5	3.5	+24.5	8	131	50/20
6	7/7	340.5	+34.5	17	14	351.5	+48.5	356	+52	6.5	90	35/5
7	6/6	349.5	+49	6.5	106	344.5	+39.5	347	+44	5	194	45/5
8	6/6	351	+36	5.5	168	346	+42.5	348	+48	5.5	169	60/6
9	6/7	302	+65	5.5	128	307	+62.5	347.5	+31	4	254	280/42
10	4/6	302	+70.5	22	10	305	+67.5	348	+34.5	13	53	277/42
11	2/2	304	+58.5	30.5	69	310	+63.5	349	+31	4	3020	280/42
12	8/8	294.5	+77.5	8.5	46	270.5	+76	353	+40	4.5	150	275/49.5
13	6/6	337	+37.5	3.5	332	338	+36	346.5	+44.5	4	259	23/13
14	3/3	299	+61.5	84	3	341	+36	341.5	+46	9	188	72/10
15	5/5	309	+72.5	10.5	54	283.5	+73	352.5	+36.5	4.5	320	277/49.5
16	6/6	330	+61	4	319	306	+64	351.5	+33.5	5.5	154	285/43
17	6/6	330.5	+63	3.5	355	328	+63.5	360	+31	3	555	289/39
18	3/3	214.5	+83	77	3	308.5	+64	351	+41.5	22	33	290/33.5
19	3/4	330	+64.5	15.5	36	327.5	+56.5	356	+33.5	6.5	365	295/32
20	4/5	323.5	+63	19.5	17	325	+50	350.5	+29	6.5	199	295/32
21	6/6	317.5	+53	9	56	312.5	+46	341	+35.5	3	511	303/28
22	0/3	Aberrant directions										
23	6/6	345	+37.5	13	29	337.5	+32	342.5	+44.5	6.5	107	45/13.5
24	7/7	347.5	+43	2	875	347.5	+43.5	348.5	+53	2	883	72/10
25	4/4	351	+37	3.5	663	350	+36	352	+46	3.5	642	69/10
26	5/5	349	+39.5	3	568	348.5	+40.5	350.5	+50	2	1307	69/10
27	5/5	351.5	+37.5	10.5	55	349.5	+35.5	351	+45	9.5	64	70/10
28	6/6	341	+40	5.5	146	335	+33.5	334.5	+45	5	182	70/12
29	4/4	339.5	+37.5	8.5	118	339	+37	339	+48.5	10.5	75	70/12
30	7/7	350	+43	13.5	20	348.5	+43	351.5	+46.5	4	232	45/5
31	8/8	352.5	+35.5	8	49	356	+30	3.5	+38.5	3.5	240	38/15
32	6/6	352	+35.5	5.5	143	350	+29	356.5	+39	4	255	38/15
33	5/5	359	+30.5	7	117	351.5	+30.5	357	+40	8	94	45/13
34	4/4	2.5	+16.5	18.5	26	360	+ 7.5	2.5	+19.5	6.5	206	55/16

No. site number; for location, see Figure 1.
R_s ratio of the number of samples used for computation to the number of samples measured.
D, I declination and inclination in degrees.
α_{95}, K semiangle of the cone of confidence of 95% probability and precision parameter, after *Fisher*'s [1953] method.
T.C. tectonic correction.
*The influence of the axial dip of about 15° has been incorporated.

TABLE 2. Late Cretaceous, Paleocene, and Eocene Data Used in This Paper

Locality	Age	D	I	α_{95}	Sites	Pole Position	Reference
		Data from Portugal					
Sintra intrusive complex	Late Cretaceous	359	+43	8	8	76.5°N, 174°E	*Van der Voo*, 1969
Monchique Nefeline-syenite	Paleocene	182	−37	14	2	73°N, 165.5°E	*Van der Voo*, 1969
Lisbon basalts	Eocene	352	+40	3	33	72.5°N, 163°W	present paper
Mean direction of magnetization for Lisbon (38.75°N, 9.25°W) computed from the three combined Portuguese results		357.5	+40.5	6.5	3		
		[351.5	+42	10	5	73.5°N, 170°W]	*Van der Voo*, 1968 (now to be discarded)
		[346.5	+37	11	12	69°N, 153W]	*Watkins and Richardson*, 1968*a* (now to be discarded)
		*Data from Stable Europe**					
Turkmenian sediments, USSR	Late Cretaceous	1.5	+19.5			61°N, 167°E	*Irving*, 1964 (pole 10.05/6)
Bulgarian andesite	Late Cretaceous	357.5	+28			66°N, 177°E	*Vollstädt et al.*, 1968
Turkmenian sediments, USSR	Paleocene, Oligocene	6.5	+18.5			60°N, 158°E	*Irving*, 1964 (pole 11.044)
British Isles, combined volcanics†	Eocene, Oligocene	4	+46			78°N, 153°E	*Irving*, 1964 (pole 11.026)
Faroe Islands	Lower Eocene	2.5	+44.5			77°N, 161°E	*Tarling*, 1970
Georgian volcanics, USSR	Eocene	344	+51			75°N, 123°W	*Irving*, 1964 (pole 11.117)
Turkmenian sediments	Oligocene	3	+34.5			70°N, 162°E	*Irving*, 1964 (pole 11.045)
Extrapolated geomagnetic field direction for Lisbon computed from the seven combined results of stable Europe		0.5	+35	10.5 (N = 7)			

* D and I computed for Lisbon.

† This combined pole position for the British Isles has been recomputed as 77°N, 171°E by *Tarling* [1970] using newer data. However, since Tarling has not yet published his reference data, we have used Irving's pole 11.026.

attempt to make paleomagnetic correlations between the six sections, comprising from south to north 6, 5, 3, 8, 5, and 3 flows (see Figure 1), failed completely. The flows apparently are of limited extent in the area. On the other hand, they appear to cover quite a considerable time span from evidence of the intercalated sediments.

CONCLUSIONS

In our opinion, the new collection has served its purpose and may settle the questions raised by the earlier studies. Summarizing the present study, we can state that, using 176 samples from 33 sites from at least some 15 flows, a mean direction of magnetization with D:352° and I:+40° was computed. This result is consistent with the mean direction given previously by *Van der Voo* [1968, 1969]: D:351.5°, I:+42°, and α_{95}:10° if unit weight is given to sites ($N = 5$), or D:354°, I:+40.5°, and α_{95}:4.5° if unit weight is given to samples in the statistical analysis ($N = 17$).

The Lisbon basalts and the two other Late Cretaceous and Eocene data from Portugal [*Van der Voo,* 1969] yield a combined geomagnetic field direction D:357.5°, I:+40.5°, and α_{95}:6.5° (Table 2). Compared with the data from stable Europe (Figure 6), though still rather unreliable, we see that there is a difference of 3° between the declination of the combined Portuguese results and the extrapolated value from the combined results for stable Europe (D:0.5°, I:+35°, and α_{95}:10.5°). This difference is not of sufficient significance to allow support for a post-Eocene rotation of the Iberian Peninsula. This evidence limits the time for practically the whole of the postulated rotation (some 35°) of the Iberian Peninsula to the period between the Triassic and Late Cretaceous–Eocene. This conclusion is not in conflict with the marine geophysical data from *Jones and Ewing* [1969], who found indications that a major part of the Bay of Biscay should have existed before Tertiary times, though they allow that an additional pre-Miocene opening of the Bay of Biscay over 12° at most might be possible. Nor is the paleomagnetic conclusion in conflict with the seismological data of the Institut Français du Pétrole [*Cholet et al.,* 1968; *Damotte et al.,* 1969; *Montadert,* 1970], which suggest Cretaceous sediments continuously deposited over the Bay of Biscay. Two holes of Leg 12 of the Deep Sea Drilling Project in the Bay of Biscay bottomed in Paleocene sediments [*Laughton et al.,* 1970].

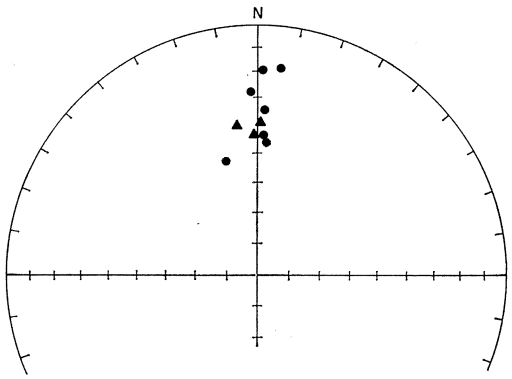


Fig. 6. Equal area projection of the mean directions of magnetization from Portuguese rocks (triangles) and extrapolated geomagnetic directions, computed for Lisbon from the pole positions obtained from rocks in stable Europe (dots). See also Table 2.

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REFERENCES

As, J. A., Instruments and measuring methods in paleomagnetic research, *Mededel. Verhandel. K.N.M.I., 78,* 56 pp., 1960.

Bullard, E. C., J. Everett, and A. G. Smith, A Symposium on Continental Drift, *Phil. Trans. Roy. Soc. London, A, 258,* 41–51, 1965.

Carey, S. W., A tectonic approach to continental drift, *Symposium on Continental Drift,* Hobart, pp. 177–355, 1958.

Cholet, J., B. Damotte, G. Grau, J. Debyser, and L. Montadert, Recherche préliminaires sur la Structure Géologique de la Marge Continentale du Golfe de Gascogne: Commentaires sur quelques Profils de Sismique Réflexion 'Flexotir,' *Rev. Inst. Fr. Pétrol. 23,* 1029–1045, 1968.

Damotte, B., J. Debyser, L. Montadert, and J. R. Delteil, Nouvelles données structurales sur le

Golfe de Gascogne obtenues par Sismique Réflexion 'Flexotir,' *Rev. Inst. Fr. Pétrol., 24,* 1061–1072, 1969.

De Jesus, A., and G. Zbyszewski, Contribution à l'étude du 'Complexe Basaltique' de Lisbonne, *Com. Serv. Geol. Portugal, 33,* 40 pp., 1952.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A, 217,* 295–305, 1953.

Girdler, R. W., Continental drift and the rotation of Spain, *Nature, 207,* 395–397, 1965.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54,* 131–167, 1949.

Irving, E., *Paleomagnetism and Its Applications to Geological and Geophysical Problems,* 399 pp., John Wiley, New York, 1964.

Jones, E. J. W., and J. I. Ewing, Age of the Bay of Biscay: Evidence from seismic profiles and bottom samples, *Science, 166,* 102–105, 1969.

Laughton, A. S., W. A. Berggren, and scientific staff, Deep Sea Drilling Project, Leg 12, *Geotimes, 15,* 10–14, 1970.

Montadert, L., Structure of the continental margin between Channel and Finisterre (abstract), *S.C.O.R. Working Group 31, Symposium 1970: Geology of the East Atlantic Continental Margin,* Cambridge, U. K., 1970.

Schwarz, E. J., A paleomagnetic investigation of Permo-Triassic redbeds and andesites from the Spanish Pyrenees, *J. Geophys. Res., 68,* 3265–3271, 1963.

Storetvedt, K. M., Paleomagnetism of the Lisbon Volcanics: A discussion of a recent paper by N. D. Watkins and A. Richardson, *Geophys. J., 19,* 107–110, 1970.

Tarling, D. H., Paleomagnetic results from the Faroe Islands, in *Palaeogeophysics,* edited by S. K. Runcorn, pp. 193–208, Academic, London, 1970.

Van der Lingen, G. J., Geology of the Spanish Pyrenees north of Canfranc, Huesca province, *Estud. Geol. Inst. Invest. Geol. 'Lucas Mallada,' Madrid, 16,* 205–242, 1960.

Van der Voo, R., The rotation of Spain: Paleomagnetic evidence from the Spanish Meseta, *Palaeogeogr., Palaeoclimatol., Palaeoecol., 3,* 393–416, 1967.

Van der Voo, R., Comments on a paper by N. D. Watkins and A. Richardson, 'The paleomagnetism of the Lisbon volcanics,' *Geophys. J., 16,* 543–547, 1968.

Van der Voo, R., Paleomagnetic evidence for the rotation of the Iberian Peninsula, *Tectonophysics, 7,* 5–56, 1969.

Van Dongen, P. G., The rotation of Spain: Paleomagnetic evidence from the eastern Pyrenees, *Palaeogeogr., Palaeoclimatol., Paleoecol., 3,* 417–432, 1967.

Vollstädt, H., K. Rother, and P. Nozharov, The paleomagnetic stability and the petrology of some Caenozoic and Cretaceous andesites of Bulgaria, *Earth Planet. Sci. Lett., 3,* 399–408, 1968.

Watkins, N. D., and A. Richardson, Paleomagnetism of the Lisbon volcanics, *Geophys. J., 15,* 287–304, 1968*a*.

Watkins, N. D., and A. Richardson, Reply by the authors to 'Comments on "Paleomagnetism of the Lisbon volcanics"' by R. Van der Voo, *Geophys. J., 16,* 549–551, 1968*b*.

Watkins, N. D., and A. Richardson, Reply by the authors to 'Paleomagnetism of the Lisbon volcanics: A discussion of a recent paper by N. D. Watkins and A. Richardson,' by K. M. Storetvedt, *Geophys. J.,* in press, 1971.

Zbyszewski, G., *Carta Geológica dos Arredores de Lisboa,* vol. 4, *Lisboa, Notica Explicativa,* Serv. Geol. Portugal, Lisbon, 93 pp., 1963.

Zbyszewski, G., *Carta Geológica dos Arredores de Lisboa,* vol. 2, *Loures, Notica Explicativa,* Serv. Geol. Portugal, Lisbon, 86 pp., 1964.

Zijderveld, J. D. A., A.C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism,* edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, Amsterdam, 1967.

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revised February 11, 1971.)

